

EDUCATION

B.Sc. in Computer Science and Engineering
Khulna University of Engineering & Technology (KUET) | Jan 2020 – Jun 2025.

SKILLS

- Programming Languages: C, C++, Python, Java.
- Web & Mobile Development: Bootstrap, Tailwind CSS, HTML, PHP, React.js, Next.js, Laravel, Flask, Android Studio, Xcode.
- Databases & Tools: MySQL, Firebase, Oracle, PL/SQL, Git
- Data Science & Machine Learning: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib.



EXPERIENCES

- Teaching Assistant at Ostad (AI Agent Development with No Code Tool for Non-coders) (August 2025 to Present)
- AI/ML Engineer – Computer Vision (Junior Level) at Transforms AI (April 2025 to September 2025)

Hajj AI System: (Built 4 computer vision systems)

Built 4 computer vision pipelines, including sentiment analysis (YOLOv11 + DeepFace), cleaner identification via uniform histogram & hashing, buffet monitoring (YOLO + VLM) for hygiene, food status, and waiter violations, and guard detection trained on a custom Ultralytics dataset.

Car Showroom Intelligence System:

Developed a face recognition system achieving ~93% accuracy to identify returning customers, implemented gender and interaction detection from CCTV footage to analyze customer-vehicle engagement, and built desk-level sentiment analysis using DeepFace, MTCNN, and FaceNet to capture real-time customer feedback.

ACADEMIC PROJECTS

- **Leveraging Mask R-CNN for Enhanced Indoor Fire Detection with a Dual-Backbone Network (Thesis):**
The research enhances indoor fire detection using Mask R-CNN with preprocessing techniques. Through transfer learning with ResNet-50 and a colour feature extractor backbone network, the model segments fire regions accurately while balancing detection accuracy and real-time performance.
- **Moodrythm** (Swift, Firebase, API) [\[github link\]](#)
 - Weather-Based Playlists: Tailors music to current weather for an immersive experience.
 - Personalized Suggestions: Curates playlists based on user preferences and atmospheric conditions.
 - Real-Time Adaptation: Dynamically updates playlists in response to weather changes with seamless, intuitive navigation.
- **Cardiac Recorder** (Java, Firebase, UI-Test, Unit-Test) [\[github link\]](#)
 - Measure heart rate and track blood pressure trends.
 - User-friendly interface with UI and unit testing for reliability.
- **Vehicle Rental APP** (Android Studio, Java, Firebase) [\[github link\]](#)
 - Developed a mobile app with secure authentication, user profile management, and vehicle booking functionality.
 - Integrated Firebase for real-time data storage and synchronization, featuring an intuitive UI with a navigation drawer for seamless user experience.
- **Vehicle Rental Website** (Laravel, Firebase, CSS) [\[github link\]](#)
 - Vehicle Booking: Rent inter-city and outer-city travel vehicles with an easy-to-use system.
 - Intuitive UI: High-quality design with user-friendly navigation.
 - Real-Time Sync: Uses Firebase for real-time data updates and management.
- **Portfolio** (React, Next-JS, Tailwind CSS) [\[link\]](#)
 - Developed a highly interactive portfolio website using React, Next.js, and Tailwind CSS to showcase skills and projects.
 - Integrated responsive design and smooth animations for an engaging user experience.

PROGRAMMING PROFILES

- Github: <https://github.com/Elin-powS>
- Portfolio: <https://swopnile-s-portfolio.vercel.app>
- Leetcode: https://leetcode.com/u/Elin_powS/
- Codeforces: https://codeforces.com/profile/elin_pows
- Kaggle: <https://www.kaggle.com/acifulsopnil>

EXTRA CURRICULAR ACTIVITIES

- Senior Executive , Spectrum(KUET)
- Senior Executive, KUET Debating Society (KDS)
- Head of Management BitFest 2.0 (Datathon)
- Achieved 4th position in SynergyX2024 Datathon.